Yagnesh Balineni

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Education

Vellore Institute of Technology (VIT) — Chennai, India
B.Tech in Electronics and Computer Engineering; CGPA: 8.01/10.0 — *2023 – 2027*

Sanskriti Junior College – Class XII (Intermediate): 97.2% — *Tirupati, India 2023*

Sri Chaitanya Techno School – Class X (SSC): 98% — *Tirupati, India 2021*

Experience

Dextora — Feb 2025 – Present
Machine Learning Intern — *Lucknow, India (Remote)*

- Develop and optimise production **computer-vision** and **deep-learning** models in **PyTorch**, and the training-to-deployment pipelines that ship them.
- Designed and built the company's **autonomous AI video-generation pipeline** (see Projects), taking a video from trend discovery to a published YouTube upload daily with **no manual steps**.

Dromos Hyperloop Team (VIT Chennai) — Jan 2025 – Present
Core Electronics Member & BMS Engineer — *Chennai, India*

- **Won Best Electrical System & Best Embedded System, GHC '26 (IIT Madras)**; Vacuum Tube Runner-Up.
- Architected a high-voltage **Battery Management System** (master–slave, **TI BQ76952 + STM32**) with **ML-based anomaly detection** for predictive cell-fault identification.
- Designed safety-critical circuits for real-time cell balancing, thermal monitoring and **CAN** communication.

VIT Chennai – Research Intern — Jun 2025 – Oct 2025
Under Dr. Sunil Kumar Pradhan — *Chennai, India*

- Built a **hybrid deep-feature-fusion** brain-tumour MRI classifier: VGG16 + Squeeze-and-Excitation attention fused with GLCM radiomics, 4-way test-time augmentation and geometric-mean stacking.
- Reached **92.44% test accuracy** on a **7,200-scan** four-class MRI dataset, with **93% glioma precision** and **100% no-tumour recall**; paper under peer review.

Projects

Autonomous AI Video Generation Pipeline | *PyTorch, Hugging Face, A100 GPUs, LLMs, FFmpeg, Whisper* — 2026

- **Fine-tuned an open-weight MiniMax image-to-video generative model** from **Hugging Face** on a **multi-GPU NVIDIA A100 cluster**, adapting its weights to turn each scene's keyframe into on-topic video.
- Engineered a fully autonomous **five-stage pipeline** (trends → LLM script → assets → edit → publish) on the **YouTube Data API** and **Claude**, publishing daily; **now used by multiple creators** to run their channels.
- Served the fine-tuned model with **ElevenLabs** neural TTS, **FFmpeg** assembly and **faster-whisper** word-level captions; **Pydantic** data contracts and a daily **GitHub Actions** run.

Energy-Efficient Spiking Neural Networks (Final-Year Thesis) | *Python, PyTorch, BrainCog* — Mar 2026 – Present

- Designed a **label-free, parameter-free early exit** for unsupervised STDP spiking networks, cutting inference energy **11.7**× (MNIST) and **16.2**× (KMNIST) at $<$**0.5 pp** accuracy loss.
- Validated on 3 datasets with bootstrap confidence intervals ($\geq$**1.61**× over the best fixed setting); lossless **4-bit synapses** and $<$**1%** neuron activity.

Deep-Shadow Volumetry | *Python, PyTorch, CNNs, Blender* — Jan 2026

- CNN pipeline reconstructing **3D volumetric shapes from 2D shadow projections**, trained on a large **synthetic dataset** generated with Blender scripting across diverse lighting conditions.

Blockchain-Based Voting System | *React.js, Solidity, Spring Boot, Web3.js* — Nov 2024

- Ethereum **smart-contract** DApp for immutable votes, with a **Spring Boot** backend, **MetaMask** authentication via Web3.js and a real-time React dashboard.

Publications

Hybrid Deep Feature Fusion and Geometric Stacking for High-Sensitivity Brain Tumor MRI Classification – B. Yagnesh, S. Tenneti. *Under peer review*, 2026.

Real-Time Object Detection Using YOLO Algorithm – peer-reviewed paper, *ICPEDC '25*, SSN College.

Technical Skills

Machine Learning: PyTorch, CNNs, Computer Vision, Transfer & Ensemble Learning, Medical Imaging, SNNs
Generative AI: Image-to-Video Fine-tuning (MiniMax), Hugging Face, Multi-GPU (A100), LLM APIs, TTS, Whisper
Languages: Python (Advanced), C, C++, Java, Verilog, SystemVerilog, Solidity
Tools & Hardware: Git, GitHub Actions, FFmpeg, Blender, MATLAB; STM32, CAN, PCB, RTL, FPGA/ASIC